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PRODUCING IN A MANUFACTURING SYSTEM WITH MINIMUM AVERAGE COST[1]

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1. INTRODUCTION

There is a considerable literature devoted to optimal production planning in continuous time manufacturing systems consisting of failure-prone machines with discounted cost criteria; see, e.g., Sethi and Zhang [7] for references and details. Not much is available, however, in connection with the long-run average cost criterion. The exceptions are Bielecki and Kumar [1] and Ghosh, Arapostathis, and Marcus [3].

In this paper, we deal with a single or parallel machine manufacturing system with convex cost of holding and backloging and develop a new rigorous analysis to address the problem. It provides the results needed for the so-called *vanishing discount approach* often used for the analysis of average cost minimization problems. In particular, we give a key lemma stating that one can go from any point in the state space to any other point in a *finite* time. The Hamilton-Jacobi-Bellman (HJB) equation is specified for the average cost problem and a verification theorem for optimality over the class of admissible controls is given. Moreover, by using the vanishing discount approach, it can be shown that the HJB equation has a viscosity solution, which turns out in this case to be also a classical solution.

The remainder of the paper is organized as follows: In Section 2 we introduce the problem and specify the required assumptions. Section 3 is devoted to the study of the value function associated with the corresponding control problem with a discounted cost. In Section 4, the Hamilton-Jacob-Bellman (HJB) equation is specified for the average cost problem and a verification theorem for optimality over the class of admissible controls is given. The analysis helps us in Section 5 to specify the optimal control policy for the average cost problem under consideration. Section 6 concludes the paper.

For proofs of the results stated in this paper, see Sethi *et al* [5].

2. PROBLEM FORMULATION

We consider a single product manufacturing system with stochastic production capacity and constant demand for its production over time. Let $x(t)$, $u(t)$, and d denote the surplus level (the

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state variable), the production rate (the control variable), and the constant demand rate, respectively. We assume $x(t) \in I\!R = (-\infty, \infty)$, $u(t) \in I\!R^+ = [0, \infty)$, $t \geq 0$, and d a positive constant. Surplus refers to inventory when $x(t) \geq 0$ and backlog when $x(t) < 0$. The system equation is

$$\dot{x}(t) = u(t) - d, \; x(0) = x. \tag{1}$$

Let $\alpha(\cdot)$ denote a Markov process with a finite state space $\mathcal{M} = \{0, 1, 2, \ldots, m\}$, with $\alpha(t)$ representing the maximum production capacity of the system at time t.

Definition 1 A production control process $u(\cdot) = \{u(t), t \geq 0\}$ is admissible if (i) $u(t) \in \mathcal{F}_t \equiv \sigma(\alpha(s), 0 \leq s \leq t)$ and (ii) $0 \leq u(t) \leq \alpha(t)$ for all $t \geq 0$.

We denote by $\mathcal{A}(k)$ the collection of admissible controls with the initial condition $\alpha(0) = k$.

Definition 2 A function $w(x, k)$ defined on $I\!R \times \mathcal{M}$ is called an admissible feedback control, or simply a feedback control, if (i) for any given initial surplus x and production capacity k, the equation

$$\dot{x}(t) = w(x(t), \alpha(t)) - d$$

has a unique solution and (ii) the control defined by $u(\cdot) = \{u(t) = w(x(t), \alpha(t)), t \geq 0\} \in \mathcal{A}(k)$.

Let $h(\cdot) : I\!R \mapsto I\!R^+$ and $c(\cdot) : [0, m] \mapsto I\!R^+$ denote the surplus (inventory/backlog) cost and the production cost, respectively. For any $u(\cdot) \in \mathcal{A}(k)$, define

$$J(x, k, u(\cdot)) = \limsup_{T \to \infty} \frac{1}{T} E \int_0^T (h(x(t)) + c(u(t)))\, dt, \tag{2}$$

where $x(\cdot)$ is the surplus process corresponding to the production process $u(\cdot)$. Our goal is to choose $u(\cdot) \in \mathcal{A}(k)$ so as to minimize the cost functional $J(x, k, u(\cdot))$.

We assume the the cost functions $h(\cdot)$, $c(\cdot)$, and the production capacity process $\alpha(\cdot)$ to satisfy the following:

(A1) $h(\cdot)$ is a nonnegative convex function with $h(0) = 0$. There are positive constants C_1, C_2, and $\kappa_0 \geq 1$ such that $h(x) \geq C_1|x|^{\kappa_0} - C_2, \quad x \in I\!R$. Moreover, there are constants $C_1' > 0$ and $\kappa \geq \kappa_0$ such that

$$|h(x) - h(y)| \leq C_1'(1 + |x|^{\kappa-1} + |y|^{\kappa-1})|x - y| \quad \forall x, y \in I\!R.$$

(A2) $c(\cdot)$ is a nonnegative twice continuously differentiable function defined on $[0, m]$ with $c(0) = 0$. Moreover, for convenience in exposition, $c(\cdot)$ is either strictly convex or linear.

(A3) $\alpha(\cdot)$ is a finite state Markov chain with generator Q, where $Q = (q_{ij})$, $i, j \in \mathcal{M}$ is a $(m+1) \times (m+1)$ matrix such that $q_{ij} \geq 0$ for $i \neq j$ and $q_{ii} = -\sum_{i \neq j} q_{ij}$. We assume that Q is *strongly irreducible* in the following sense: the equations

$$\nu Q = 0 \quad \text{and} \quad \sum_{i=0}^{m} \nu_i = 1$$

have a unique solution $\nu = (\nu_0, \nu_1, \ldots, \nu_m)$ with $\nu_k > 0$, $k = 0, 1, \ldots, m$.

(A4) The average capacity $\bar{\alpha} \equiv \sum_{i=0}^{m} i\nu_i > d$.

(A5) $d \notin \mathcal{M}$.

A control $u(\cdot) \in \mathcal{A}(k)$ is called *stable* if it satisfies the condition

$$\lim_{T\to\infty} \frac{E|x(T)|^{\kappa+1}}{T} = 0, \tag{3}$$

where $x(\cdot)$ is the surplus process corresponding to the control $u(\cdot)$ with $(x(0), \alpha(0)) = (x, k)$ and κ is defined in Assumption (A1). Let $\mathcal{B}(k) \subset \mathcal{A}(k)$ denote the class of stable controls.

We will show in Section 4 that there exists a constant λ^*, independent of the initial condition $(x(0), \alpha(0)) = (x, k)$, and a stable Markov control policy $u^*(\cdot) \in \mathcal{A}(k)$ such that $u^*(\cdot)$ is optimal, i.e., it minimizes the cost defined by (2) over all $u(\cdot) \in \mathcal{A}(k)$, and furthermore,

$$\lim_{T\to\infty} \frac{1}{T} E \int_0^T (h(x^*(t)) + c(u^*(t)))\, dt = \lambda^*,$$

where $x^*(\cdot)$ is the surplus process corresponding to $u^*(\cdot)$ with $(x(0), \alpha(0)) = (x, k)$. Moreover, for any other (stable) control $u(\cdot) \in \mathcal{B}(k)$,

$$\liminf_{T\to\infty} \frac{1}{T} E \int_0^T (h(x(t)) + c(u(t)))\, dt \geq \lambda^*.$$

3. ANALYSIS OF THE DISCOUNTED COST PROBLEM

We introduce a corresponding control problem with the cost discounted at a rate $\rho > 0$. For $u(\cdot) \in \mathcal{A}(k)$, we define the expected discounted cost as

$$J^\rho(x, k, u(\cdot)) = E \int_0^\infty \exp(-\rho t)(h(x(t)) + c(u(t)))\, dt.$$

Define the value function of the discounted cost problem as

$$V^\rho(x, k) = \inf_{u(\cdot)\in\mathcal{A}(k)} J^\rho(x, k, u(\cdot)). \tag{4}$$

The HJB equation associated with this problem is

$$\rho V^\rho(x, k) = F(k, V_x^\rho(x, k)) + h(x) + QV^\rho(x, \cdot)(k), \tag{5}$$

where $V_x^\rho(\cdot, \cdot)$ is the partial derivative of $V^\rho(\cdot, \cdot)$ with respect to its first variable,

$$F(k, r) = \inf_{0\leq u\leq k}((u - d)r + c(u)),$$

and the operator Q is defined by

$$QV^\rho(x, \cdot)(k) = \sum_{k'\neq k} q_{kk'}(V^\rho(x, k') - V^\rho(x, k)). \tag{6}$$

Theorem 1 (Sethi *et al.* [4]) The value function V^ρ has the following properties:

(i) $V^\rho(\cdot, k)$ is continuously differentiable and convex for any fixed $k \in \mathcal{M}$. Moreover, there are positive constants C_1, C_2, and C_3 such that for any k

$$C_1|x|^{\kappa_0} - C_2 \leq V^\rho(x, k) \leq C_3(1 + |x|^\kappa),$$

where κ, κ_0 are as defined in Assumption (A1).

(ii) $V^\rho(\cdot,\cdot)$ is the unique solution of the HJB equation (5).

In order to study the long-run average cost control problem using the vanishing discount approach, we must first obtain some estimates for the value function $V^\rho(x,k)$.

Lemma 1 There exists a constant $C > 0$ such that for any $(x_0,k_0),(x,k) \in I\!R \times \mathcal{M}$, we can find an admissible control $u(\cdot) \in \mathcal{A}(k_0)$ such that for $r \geq 1$

$$E\tau^r \leq C(1 + |x - x_0|^r) \quad , \tag{7}$$

where

$$\tau \equiv \inf\{t > 0 : (x(t), \alpha(t)) = (x,k)\}, \tag{8}$$

and $x(\cdot)$ is the surplus process corresponding to the control policy $u(\cdot)$ and the initial condition $(x(0),\alpha(0)) = (x_0,k_0)$.

The following theorem can be shown by using Lemma 1 and the dynamic programming principle.

Theorem 2 There exists a constant $\rho_0 > 0$ such that $\rho V^\rho(0,0)$ is bounded for $0 < \rho \leq \rho_0$.

Let us define the differential discounted value function

$$\bar{V}^\rho(x,k) = V^\rho(x,k) - V^\rho(0,0), \tag{9}$$

for which the following results can be derived.

Theorem 3 The function $\bar{V}^\rho(x,k)$ is convex in x. It is locally uniformly bounded, i.e., there exists a constant $C > 0$ such that

$$|\bar{V}^\rho(x,k)| \leq C(1 + |x|^{\kappa+1}) \quad \forall (x,k) \in I\!R \times \mathcal{M}, \rho \geq 0. \tag{10}$$

The following two results follow easily from Theorem 3

Corollary 1 $\bar{V}^\rho(x,k)$ is locally uniformly Lipschitz continuous in x with respect to $\rho > 0$, i..e., for any $X > 0$, there exists a constant $C > 0$, independent of ρ, such that

$$|\bar{V}^\rho(x,k) - \bar{V}^\rho(x',k)| \leq C|x - x'|$$

for all $k \in \mathcal{M}$ and all $|x| \leq X$ and $|x'| \leq X$.

Corollary 2 For $(x,k) \in I\!R \times \mathcal{M}$, there is a subsequence of ρ, still denoted by ρ, such that

$$\lambda^* = \lim_{\rho \to 0} \rho V^\rho(x,k) \quad \text{and} \quad V(x,k) = \lim_{\rho \to 0} \bar{V}^\rho(x,k). \tag{11}$$

Moreover, the convergence is locally uniform in (x,k) and $V(\cdot)$ is locally Lipschitz continuous.

4. VERIFICATION THEOREM

The HJB equation associated with the long-run average cost optimal control problem formulated in Section 3 takes the following form:

$$\lambda = F(k, W_x(x,k)) + h(x) + QW(x,\cdot)(k), \tag{12}$$

where λ is a constant and W is a real-valued function defined on $\mathbb{R} \times \mathcal{M}$.

Before we define a solution to the HJB equation (12), we first introduce some notation. Let $\mathcal{G}$ denote the family of real-valued functions $W(\cdot,\cdot)$ defined on $\mathbb{R} \times \mathcal{M}$ such that (i) $W(\cdot,k)$ is convex; (ii) $W(\cdot,k)$ is continuously differentiable; (iii) $W(\cdot,k)$ has polynomial growth, i.e., there are constants $l, C > 0$ such that

$$|W(x,k)| \le C(1+|x|^l) \qquad \forall x \in \mathbb{R}.$$

A solution to the HJB equation (12) is a pair (λ, W) with λ a constant and $W \in \mathcal{G}$. The function W is called a *potential function* for the control problem, if λ is the minimum long-run average cost.

We now show that the limits (λ^*, V) defined by (11) in Corollary 2 is indeed a solution to the HJB equation (12). Since the HJB equation involves the partial derivative $V_x(\cdot,\cdot)$, which may not exist for the defined function $V(\cdot,\cdot)$, we must first interpret the solution of the HJB equation (12) in the *viscosity sense*. This is done in Theorem 4. In Theorem 5 we show that $V(\cdot,k)$ is continuously differentiable, and therefore (λ^*, V) is indeed a classical solution. We refer to Fleming and Soner [2] for the definition of viscosity solutions and some relevant results.

Theorem 4 (λ^*, V) is a viscosity solution to the HJB equation (12). Moreover, the constant λ^* is unique in the following sense: If (λ, V') is another viscosity solution to (12), then $\lambda = \lambda^*$.

Theorem 5 The function $V(x,k)$ is continuously differentiable in x, and (λ^*, V) is a classical solution to the HJB equation. Moreover, $V(x,k)$ is convex in x and

$$|V(x,k)| \le C(1+|x|^{\kappa+1}).$$

The following verification theorem can be stated now.

Theorem 6 Let (λ, W) be a solution to the HJB equation (12). Then

(i) If there is a control $u^*(\cdot) \in \mathcal{A}(k)$ such that

$$F\left(\alpha(t), W_x(x^*(t),\alpha(t))\right) = (u^*(t)-d)W_x(x^*(t),\alpha(t)) + c(u^*(t)) \tag{13}$$

for a.e. $t \ge 0$ with probability 1, where $x^*(\cdot)$ is the surplus process corresponding to the control $u^*(\cdot)$, and

$$\lim_{T\to\infty} \frac{W(x^*(T),\alpha(T))}{T} = 0, \tag{14}$$

then

$$\lambda = J(x,k,u^*(\cdot)).$$

(ii) For any $u(\cdot) \in \mathcal{A}(k)$, we have $\lambda \le J(x,k,u(\cdot))$, i.e.,

$$\limsup_{t\to\infty} E \int_0^t (h(x(t)) + c(u(t)))\, dt \ge \lambda.$$

(iii) Furthermore, for any (stable) control policy $u(\cdot) \in \mathcal{B}(k)$, we have

$$\liminf_{t\to\infty} \frac{1}{t} E \int_0^t (h(x(t)) + c(u(t)))\, dt \ge \lambda. \tag{15}$$

5. CHARACTERIZATION OF OPTIMAL CONTROL

From Theorem 5, we know that the function $V \in \mathcal{G}$, and that it is also a solution of the HJB equation (24). The function V is sometimes referred to as the relative value function. Let us now define a control policy $u^*(\cdot,\cdot)$ via the relative function $V(\cdot,\cdot)$ as follows:

$$u^*(x,k) = \begin{cases} 0 & \text{if} \quad V_x(x,k) > -c_u(0) \\ (c_u)^{-1}(-V_x(x,k)) & \text{if} \quad -c_u(k) \le V_x(x,k) \le -c_u(0) \\ k & \text{if} \quad V_x(x,k) < -c_u(k) \end{cases} \tag{16}$$

if the function $c(\cdot)$ is strictly convex, or

$$u^*(x,k) = \begin{cases} 0 & \text{if} \quad V_x(x,k) > -c \\ k \wedge d & \text{if} \quad V_x(x,k) = -c \\ k & \text{if} \quad V_x(x,k) < c \end{cases} \tag{17}$$

if $c(u) = cu$. Therefore, the control policy $u^*(\cdot,\cdot)$ satisfies the condition (13).

From the convexity of the function $V(\cdot,k)$, there are x_k, y_k, $-\infty < y_k < x_k < \infty$ such that

$$U(x) = (x_k, \infty) \quad \text{and} \quad L(k) = (-\infty, y_k).$$

The control policy $u^*(\cdot,\cdot)$ can be written as

$$u^*(x,k) = \begin{cases} 0 & x > x_k, \\ (c_u)^{-1}(-V_x(x,k)) & y_k \le x \le x_k, \\ k & x < y_k. \end{cases}$$

Now we are in the position to state and prove the following theorem.

Theorem 7 The control policy $u^*(\cdot,\cdot)$, defined in (16) or (17) as the case may be, is optimal.

Proof. By Theorem 6, we need only to show that

$$\lim_{t\to\infty} \frac{V(x^*(t), \alpha(t))}{t} = 0.$$

But, this is implied by Theorem 5 and the fact that $u^*(\cdot,\cdot)$ is a stable control.

Remark 1 When $c(u) = 0$, i.e., there is no production cost in the model, the optimal control policy can be chosen to be the so-called *hedging point policy*, which has the following form: There are real numbers x_k, $k = 1, \ldots, m$, such that

$$u^*(x,k) = \begin{cases} 0 & x > x_k \\ k \wedge d & x = x_k \\ k & x < x_k. \end{cases}$$

6. CONCLUDING REMARKS

In this paper, we have used the vanishing discount approach to develop the theory of dynamic programming for single or parallel machines, convex cost stochastic manufacturing systems with the long-run average cost minimization criterion. We have generalized previous results as well as provided a theoretical framework for various heuristic analyses of such systems in the literature. A multiproduct extension of this paper uses an HJB equation in terms of the directional derivative, see [6].

Further research should focus on extending the analysis to more complex manufacturing systems such as flowshops and jobshops. For such models with discounted cost criteria, see Sethi and Zhang [7].

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